pfi

Python for Fixed Income

COURSE 10 · IMPLEMENTING RAG

A Filing as Pages and Items

WEEK 1 · DAY 2

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Today: a filing as pages and items

One filing, two copies

PDF	the course's print, 113 pages
HTML	the 10-K as filed (Course 7)
Pages	what a reader opens and cites
Items	Item 1 to Item 16; never crossed
Model	none today; nothing is sent

Today

1. Why a PDF
2. pdf_pages: the pages
3. Four pages read as text
4. Pages against the HTML
5. page_items: the items
6. Printed pages against PDF pages
7. table_like_lines; a debt table

Look at the pages before anything is chunked: a chunker sees only what they hold.

Why load a PDF when the HTML is there

Two reasons from the desk

Most documents reach a desk as PDFs: offering memoranda, indentures, agency reports.

Pages are what a reader cites: page 37 of a stated file can be opened and checked.

7,729,841

bytes in the PDF: fonts and line positions

4,008,734

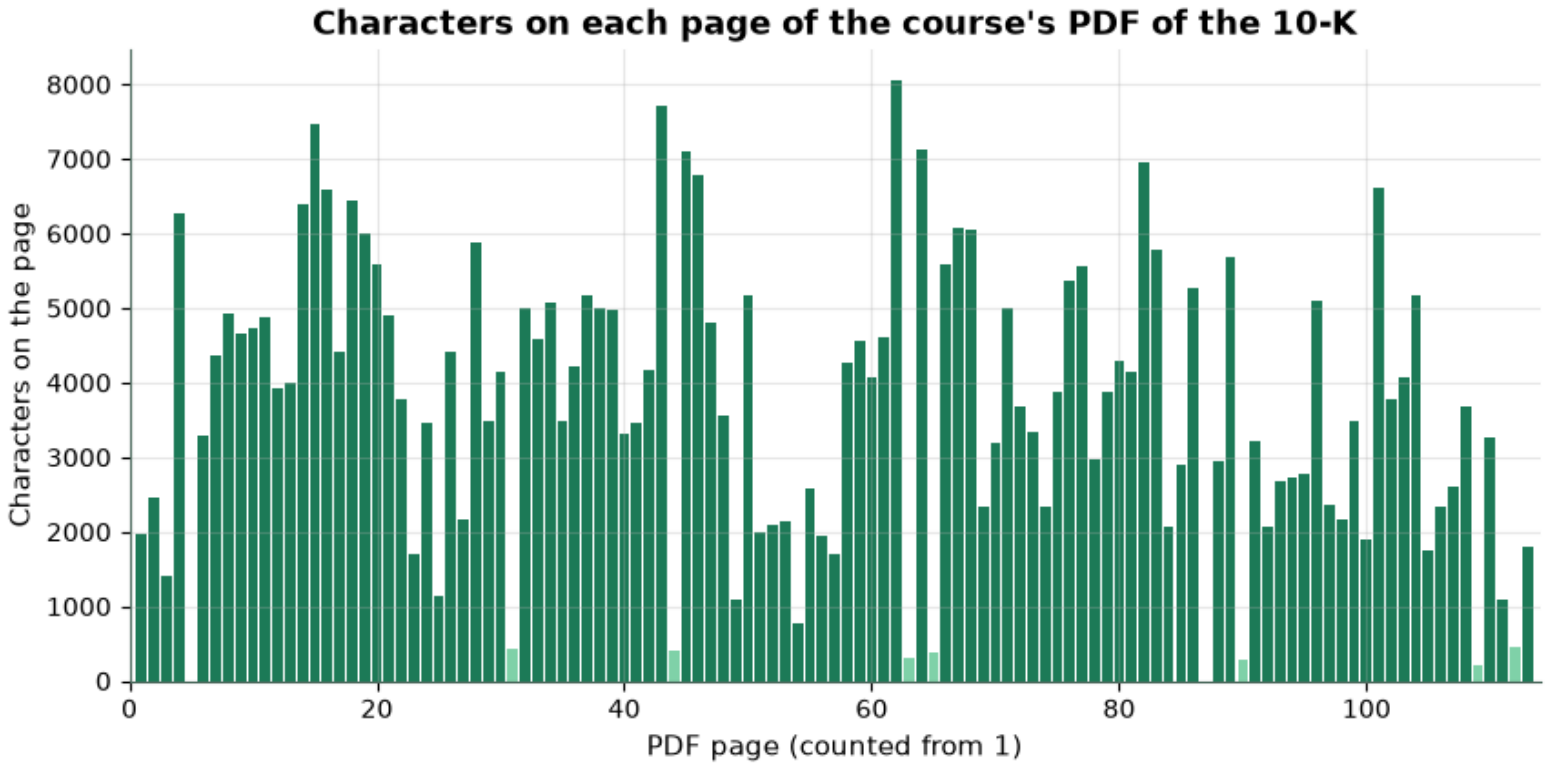
bytes in the HTML as filed

422,118

characters of HTML text, tags removed

EDGAR lists no PDF of this 10-K: the course printed one, and cites its pages.

The PDF as pages



pdf_pages found

113
pages

421,643
characters

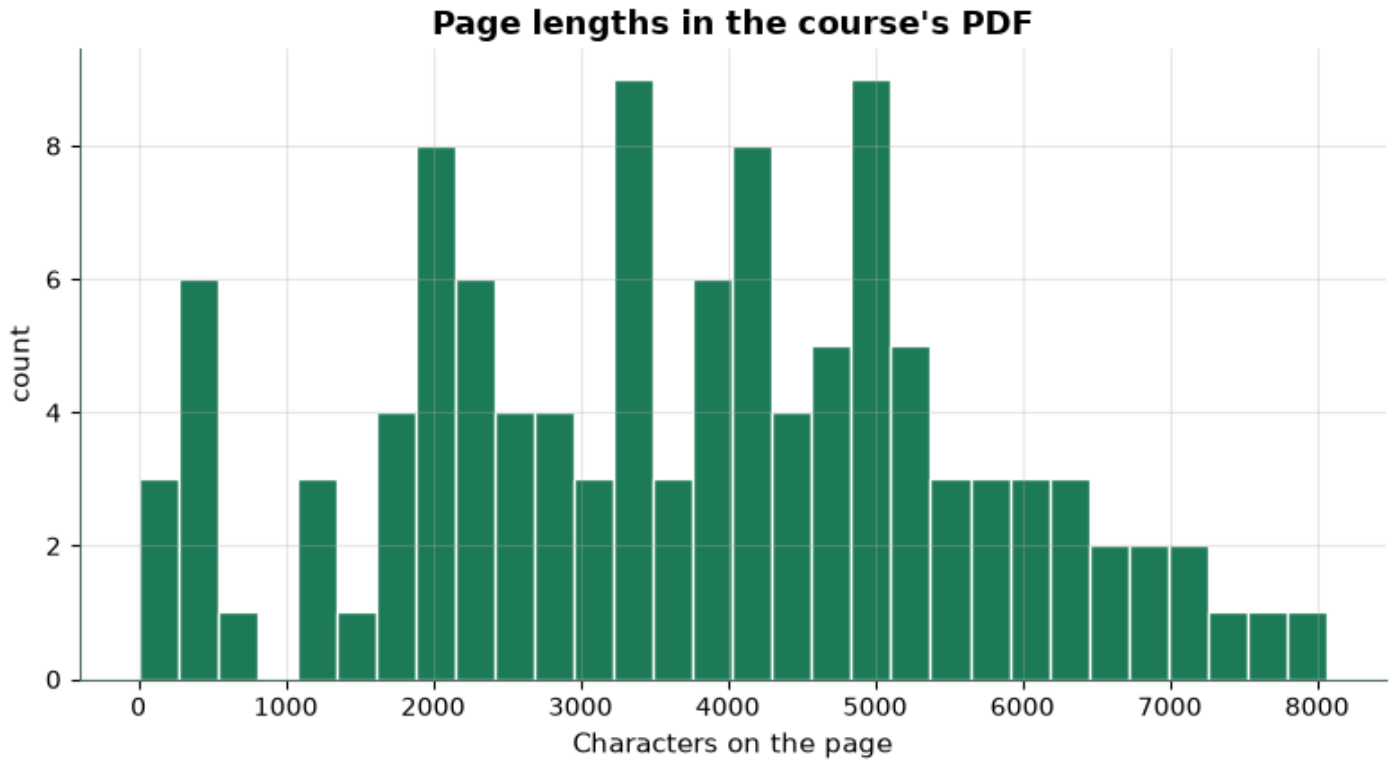
3,731
mean per page

page 87
empty, kept

8 pages
under 500 characters

102
end with a printed number

Page lengths



A typical page

3,775 characters

a median page; day 3's chunks are at most 512 characters each

The empty page

Page 87 has no text layer. Kept as a row of 0 characters, so page 88 stays 88.

Reading pages as text

p. 1 The cover

form text, box characters, and an image's file name

p. 3 The contents

items with the 10-K's printed page numbers

p. 38 Item 7 prose

line breaks where the print put them

p. 74 The debt table

Note 7: rows intact, column heads broken

105 of 113

pages start with the line "Table of Contents"

7

short sheets a printed page ran onto

page 87

empty: no header, no text

A chunker sees exactly what pdf_pages returns, running header included.

Pages against the HTML

```
squeeze = lower case, letters and digits only
found = squeeze(page) in squeeze(html_text)
```

111 of 113

pages found in the HTML, letter for letter

page 1

holds "logoboyd.jpg": the logo's file name

page 25

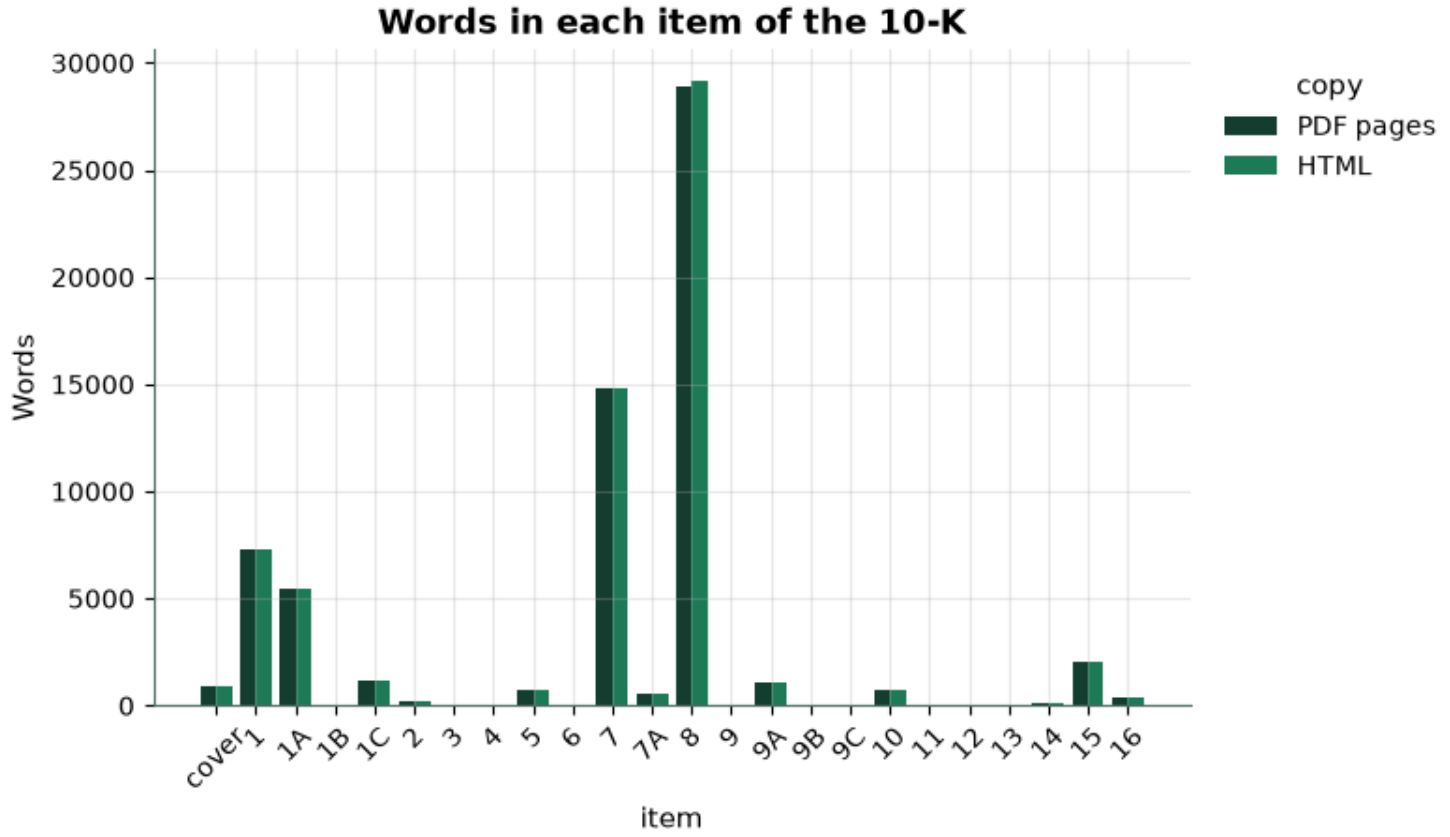
holds "byd2025.jpg": a graph's file name

An empty page passes trivially: an empty text is inside any text.

So the check proves the PDF adds nothing; it cannot prove the PDF dropped nothing.

The PDF's text is the filing's text; the print moved line breaks and page breaks.

Items: where each section starts and ends



page_items

23

items found, none missing

136

rows from 113 pages

13

pages that hold two items or more

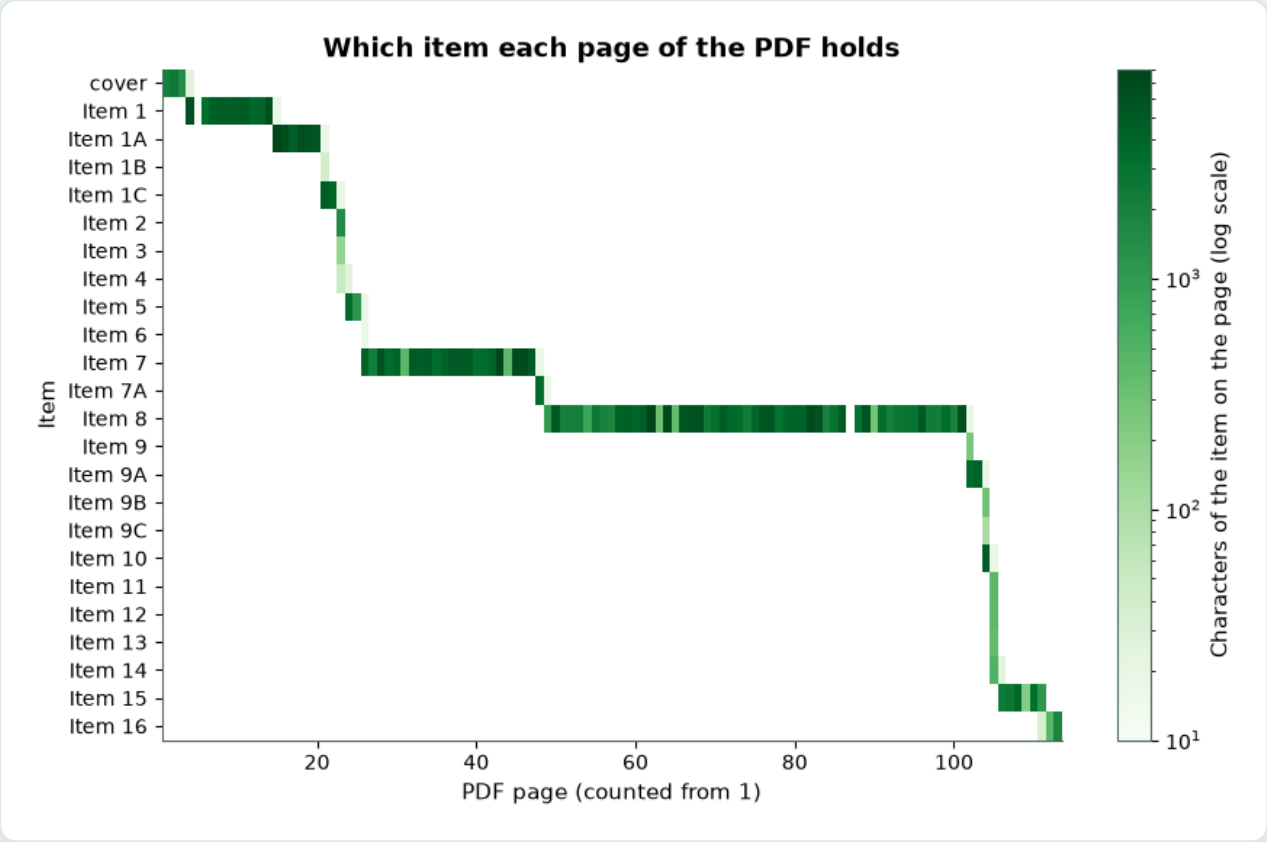
5 on p. 105

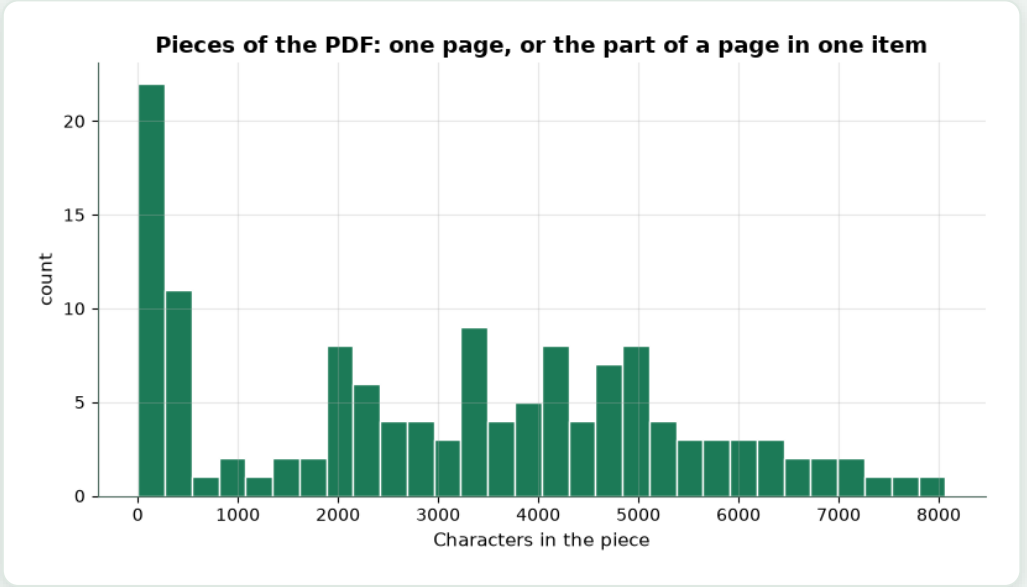
the Part III items

never

a chunk across two items

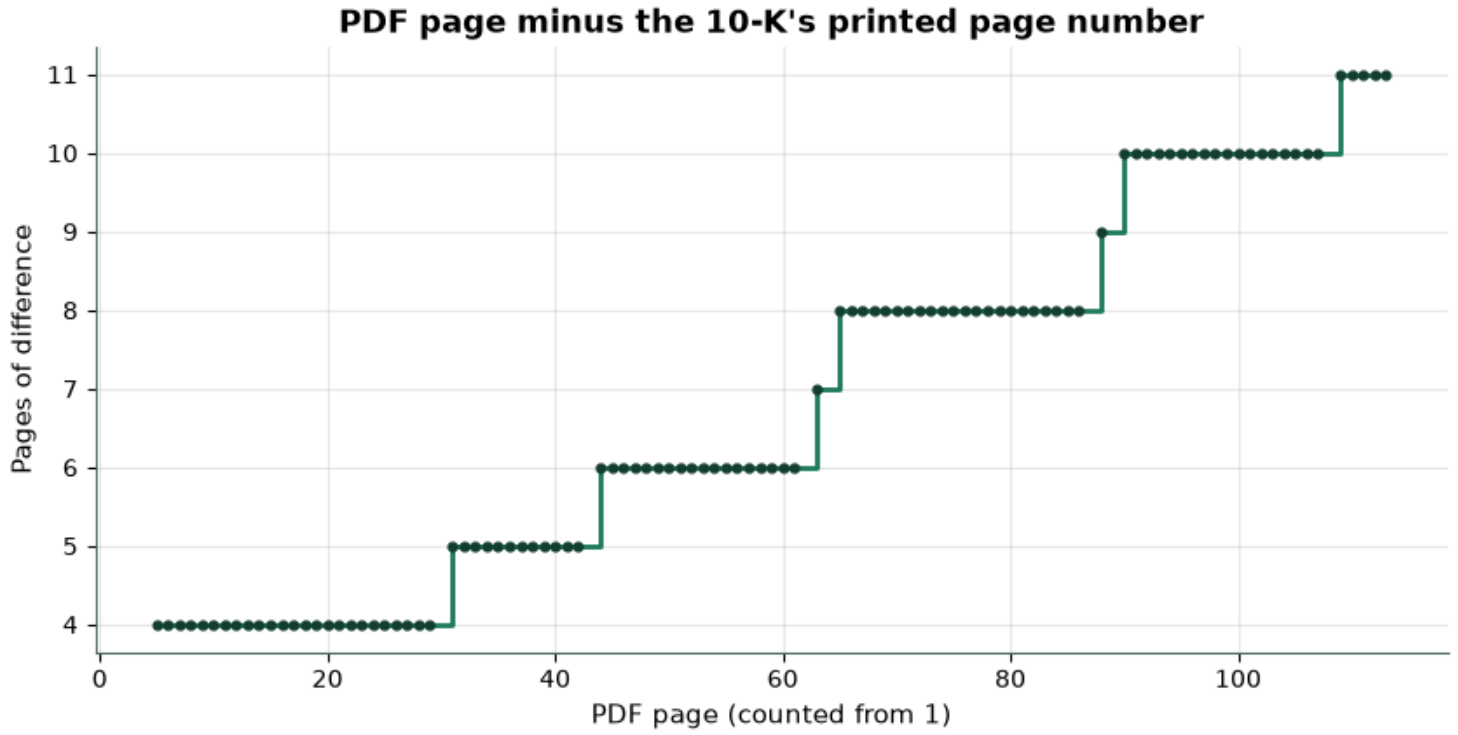
A page-by-item map, and the pieces a chunker sees





136 pieces; 18 under 100 characters; 9 only "Table of Contents".

Printed page numbers against PDF pages



What the pages show

23 of 23

headings at the contents' page

+4

on the first numbered page

+11

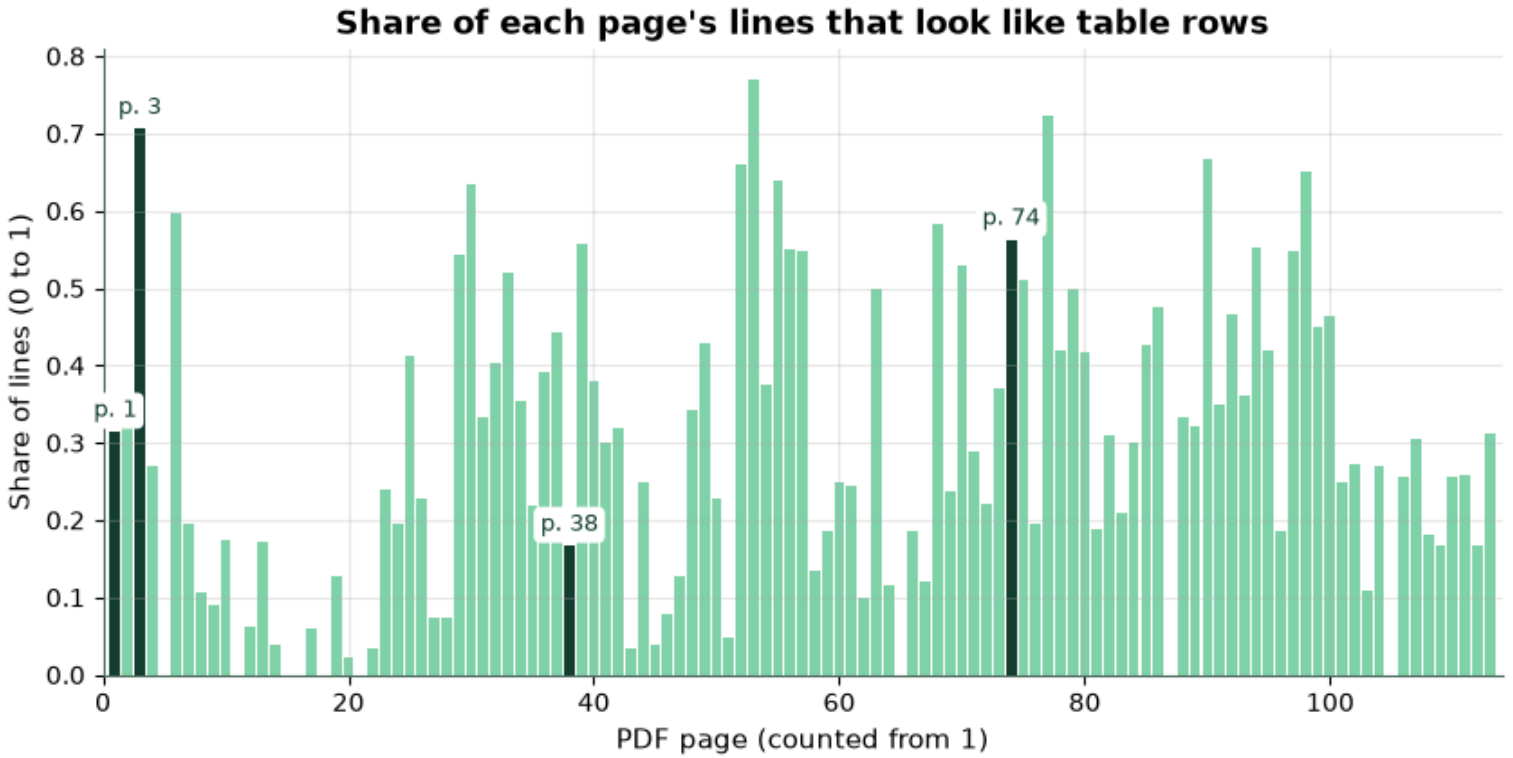
on the last

1 to 102

the 10-K's printed pages

The course cites the PDF's page: the one anyone holding the file can open.

Table-like lines on each page



The four pages read

p. 1 cover	**0.3143**
p. 3 contents	**0.7059**
p. 38 Item 7	**0.1667**
p. 74 debt table	**0.5625**

Mean 0.2929; 20 pages over one half; 0 lines with a run of three spaces.

What a text layer does to a debt table

Page 74, from Note 7 (quoted as published)

```
NOTE 7. LONG-TERM DEBT

Long-term debt, net of current maturities and debt issuance cos...

December 31, 2025

Interest Unamortized

Rates at Origination

December 31, Outstanding Fees and Long-Term

(In thousands) 2025 Principal Costs Debt, Net

Credit facility 5.318% $ 160,700 $ (3,820) $ 156,880

4.750% senior notes due 2027 4.750% 1,000,000 (3,896) 996,104

4.750% senior notes due 2031 4.750% 900,000 (7,415) 892,585

Long-term debt, net $ 2,060,700 $ (15,131) $ 2,045,569
```

Page 75, its top

```
Table of Contents

BOYD GAMING CORPORATION AND SUBSIDI...

NOTES TO CONSOLIDATED FINANCIAL STA...

as of December 31, 2025 and 2024 an...

December 31, 2024

Interest Unamortized

Rates at Origination

December 31, Outstanding Fees and L...
```

The 2024 block starts here; "NOTE 7" is on page 74 only.

Source: Boyd Gaming Corporation, Form 10-K for fiscal 2025, filed 2026-02-20, accession 0001437749-26-004908, Item 8, page 74 of the course's PDF.

Excel side by side, and Git

Excel side by side (tested in Excel)

`=LEN(A2), filled down (Text cells)`	equal on 48 of 48 lines
`=SUM(B2:B49)`	2,299, equal to Python
`=COUNTA(A2:A49)`	48 lines, equal
`"December 31, 2025" in a General cell`	a date: 46022; LEN 5, not 17

Format the column as Text before you paste a page. An item over 32,767 characters needs several cells.

Git: one commit of nine files: the modules, ragkit with 14 tests, .gitignore, and pages.csv.

The AI check

The docstring is the prompt. The retrieval code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: an empty page is skipped, so every later page is numbered one early.

Code that runs is not the same as code that is right.

What you can do now

- Say why a desk loads a PDF, and cites its pages
- Read every page with pdf_pages, the empty one kept
- Check a PDF against the HTML; say what that proves
- Place pages in items with page_items; cite the PDF's page
- Measure table rows; say what a text layer loses

Week 1, Day 3 Chunking: size and overlap